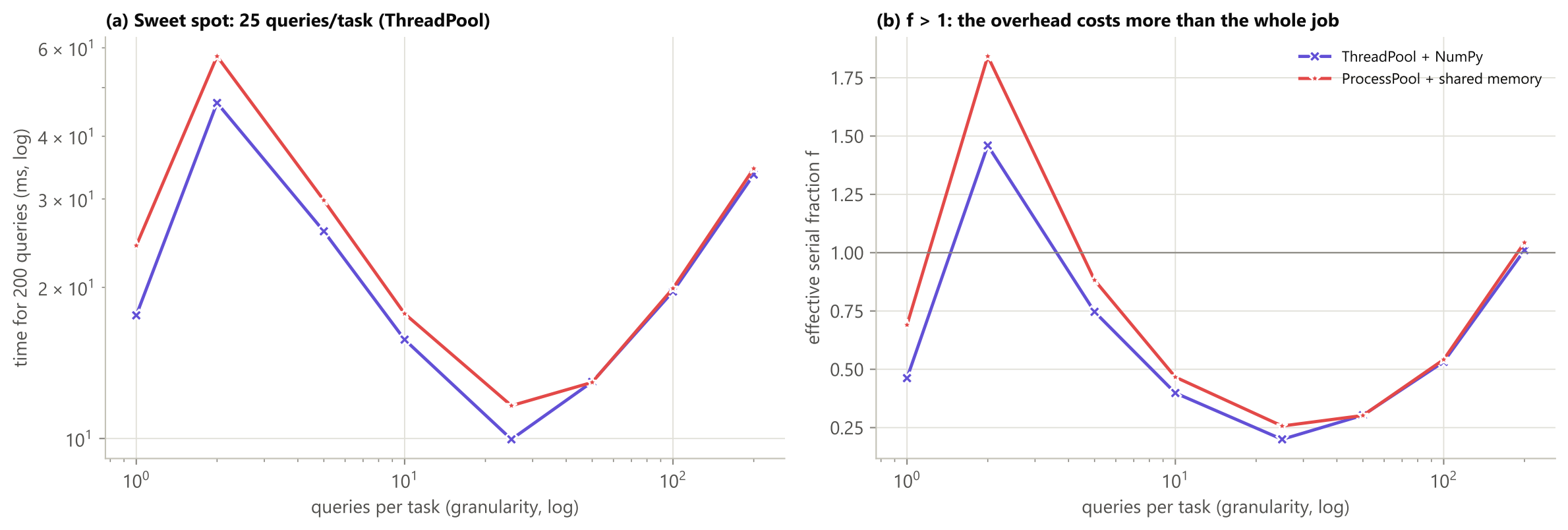
(a) Sweet spot: 25 queries/task (ThreadPool)
time for 200 queries (ms, log)
6 × 10^1
4 × 10^1
3 × 10^1
2 × 10^1
10^1
10^0
10^1
10^2
queries per task (granularity, log)
(b) f > 1: the overhead costs more than the whole job
effective serial fraction f
1.75
1.50
1.25
1.00
0.75
0.50
0.25
ThreadPool + NumPy
ProcessPool + shared memory
10^0
10^1
10^2
queries per task (granularity, log)